

Nguyen Minh Khoa

AI Engineer & Researcher

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I'm Khoa Nguyen, an AI engineer and Research Assistant at AiTA Lab, FPT University.

My recent work includes handwritten mathematical expression recognition and text-to-molecule generation. I have worked across data preparation, model experiments, and inference integration, with a preference for research that can be checked in working software.

I'm now focusing on AI for manufacturing, especially visual systems that must cope with limited data and changing operating conditions. I'm looking for work that gives equal weight to careful experiments and hands-on implementation.

I'm open to AI engineering, research engineering, and applied machine-learning collaborations.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University | Oct 2024 - Present

- First-authored LexiChem, accepted at APWeb-WAIM 2026, and presented it in Da Nang. Developed method ideas, ran experiments and ablations, wrote the Introduction and Results & Discussion, and co-wrote Methods.
- Co-first-authored Mask CoMER (ICDAR 2025) and co-authored the MTL + CBARM journal paper, contributing pretraining and multi-task learning experiments for mathematical expression recognition.
- Co-first-authored ChemAligner-T5 (ICCIES 2026), a study aligning text and molecular representations for text-to-molecule generation.
- Reviewed and refined the methodology for CorrTie, an active-learning paper accepted at APWeb-WAIM 2026, as a co-author.

RESEARCH & ENGINEERING PROJECTS

LexiChem | AI & Inference Contributor | 2025 - 2026

Completed capstone | Three-person team | Local application

- Built the AI path in a three-person capstone, bringing text-to-molecule models into an application where users generate, compare and inspect molecular candidates in 2D/3D.
- Created training-ready SELFIES data from Mol-Instructions, L+M-24 and ChEBI-20 by validating structures, canonicalizing SMILES and removing duplicates with RDKit.
- Developed shared-latent text/SELFIES alignment ideas and tested them through experiments and ablations, forming the research core of the first-author APWeb-WAIM paper.
- Integrated FastAPI/Triton inference, RDKit structure checks and MongoDB storage into Experiments, connecting prompt entry to validated molecules and saved runs.

Tools: Python, PyTorch, Transformers, RDKit, SMILES/SELFIES, FastAPI, NVIDIA Triton, MongoDB

NexOps | AI, Simulation & Platform Lead | 2025 - Present

MES/SCADA working prototype | Local factory-operations runtime

- Built the local runtime linking machine telemetry, work orders, production plans, alarms and OEE across operator views, leading the project's AI, simulation and platform work.
- Created repeatable machine and shift simulations to exercise planning and alarm workflows under the same factory scenarios across runs.
- Connected simulator and edge events to live dashboards through MQTT/EMQX, FastAPI and TimescaleDB, making machine state and shift metrics visible in one workspace.
- Integrated a Bambu Lab printer over LAN and demonstrated device control in a recorded trial, extending the prototype beyond simulated machines.

Tools: Python, FastAPI, MQTT/EMQX, PostgreSQL/TimescaleDB, React, TypeScript, Docker Compose

RESEARCH & ENGINEERING PROJECTS - CONTINUED

HMER | Research Contributor | 2024 - 2026

Handwritten mathematical expressions to LaTeX | AiTA Lab

- Co-first-authored Mask CoMER, published in the ICDAR proceedings, with experimental contributions to masked-language pretraining for handwritten expression recognition.
- Co-authored the MTL + CBARM paper in the Journal of Information and Telecommunication through multi-task learning experiments combining structural supervision and attention refinement.

Tools: Python, PyTorch, PyTorch Lightning, CUDA, Weights & Biases, CROHME

Research Ops | Developer | 2026 - Present

Personal project | Pre-alpha research infrastructure

- Built a resumable research workspace that preserves the sources, evidence and decisions behind each revision, keeping research context available across interrupted sessions.
- Implemented revision snapshots and idempotent commands in FastAPI/PostgreSQL to preserve prior state and handle repeated requests, with live updates delivered through SSE.
- Linked PDF evidence to source-file digests and manuscript exports to specific revisions, making the source of a citation and the state behind an export traceable.
- Verified the local document workflow from PDF import and evidence-span checking through manuscript export and compilation; kept runtime changes behind human approval.

Tools: Python, FastAPI, PostgreSQL, TypeScript, React, Prefect

PUBLICATIONS

Published as **Khoa Minh Nguyen**.

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | First author | Accepted & presented

Developed method ideas and ablations; wrote the Introduction and Results & Discussion, and co-wrote Methods. Presented at APWeb-WAIM in Da Nang.

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pretraining and Regularization

ICDAR 2025 | Co-first author | Published; proceedings 2026

Experimental contributions to masked-language pretraining for HMER. LNCS 16025, pp. 375-390.

ChemAligner-T5: A Unified Text-to-Molecule Model via Representation Alignment

ICCIES 2026 | Co-first author | Published

Co-first-author work on aligning text and molecular representations with BioT5+, evaluated on L+M-24.

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | Co-author | Published online 19 August 2026

Experimental contributions to multi-task learning for HMER with structural supervision and attention refinement.

CorrTie: Correction-aware Tie-breaking for Active Learning with Vision-Language Models

APWeb-WAIM 2026 | Co-author | Accepted

Critical review and methodological refinement of correction-aware sample selection for vision-language active learning.

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026 | Academic requirements completed; degree conferral expected November 2026.

Final GPA: 8.0/10.

AWARDS & SCHOLARSHIPS

50% Tuition Scholarship | FPT University | 2022

Earned through the university scholarship examination, covering half of tuition for the full programme.

Second Prize | RESFES HCMC | 2026

Awarded to the ChemAligner-T5 team at FPT University HCMC's Spring 2026 student research competition.

Second Prize | RESFES HCMC | 2025

Awarded to the Mask CoMER team at FPT University HCMC's student research competition.

VND 50 Million Team Award & Funding | Demo Pitching Day | 2025

TrendRadar team recognition and project funding from FPT University.

VND 40 Million Team Support | Innovation Quest | 2025

TrendRadar, FrontAI team, through SIHUB; includes a VND 10 million team reward.

Silver Medal | HCMC Open April Olympiad | 2021

Mathematics, Grade 11.

TECHNICAL SKILLS

AI & research: Python, PyTorch, Transformers, model experiments and ablations, HMER, cross-modal learning, text-to-molecule generation

Systems & data: FastAPI, NVIDIA Triton, RDKit, PostgreSQL/TimescaleDB, MongoDB, MQTT, Docker, Git

CERTIFICATIONS

AIO | AI VIET NAM: Machine Learning; Deep Learning; Computer Vision & NLP; GenAI & LLMs.

Machine Learning | DeepLearning.AI · Stanford Online | 2023

Deep Learning | DeepLearning.AI | 2024

Natural Language Processing | DeepLearning.AI | 2025

IBM Full Stack Software Developer | IBM | 2024

Additional AI training: DeepLearning.AI TensorFlow Developer; Big Data; Ethics in the Age of AI; AI Foundations for Everyone; Introduction to AI.

Software & cloud: AWS Cloud Technical Essentials; Software Development Lifecycle.

Research & academic writing: Research Methodologies; Being a researcher (in Information Science and Technology); Understanding Research Methods; Introduction to Research for Essay Writing; Advanced Writing; Academic Skills for University Success.

Project management: Project Management Principles and Practices.